

E-COMMERCE ORDER SALES ANALYSIS

Using Python and Power BI

Data Analytics Final Project Document

Domain: Sales & E-commerce | Retail Analytics

Tools Used: Python (Pandas, Matplotlib, Seaborn) + Power BI

Dataset Source: Hugging Face - millat/e-commerce-orders

1. Project Overview and Objective

This project involves cleaning, transforming, and analysing raw e-commerce order data using Python (Pandas, NumPy, Matplotlib, and Seaborn) and creating two interactive Power BI dashboards to derive meaningful business insights.

The main objective is to demonstrate end-to-end data pre-processing techniques using Python along with interactive Power BI dashboard visualizations, in order to make informed, data-driven decisions about sales performance, customer behaviour, and delivery operations.

2. Data Sources

- Source: Hugging Face - millat/e-commerce-orders (huggingface.co/datasets/millat/e-commerce-orders)
- DOI / License: 10.57967/hf/5258 | MIT License
- Timeline: Orders placed between April 2024 and April 2025
- Records Used: A random sample of 1000 rows from the full 10,000-row dataset, 15 original columns
- Domain: Sales & E-commerce / Retail Analytics

3. Problem Statement

- To identify meaningful patterns, trends, and insights from raw e-commerce order data using exploratory data analysis (EDA).
- To clean, transform, and preprocess messy real-world order data - handling missing values, duplicates, and outliers.
- To understand which product categories and customer segments drive the most revenue.
- To evaluate delivery performance, and marketing channel / device effectiveness, to support operational and marketing decisions.

4. Attribute (Column / Features) Details

The raw dataset contains the following 15 columns before cleaning:

Attribute Name	Data Type	Description
order_id	String (UUID)	Unique identifier for each order
customer_id	Integer	Unique identifier for the customer who placed the order
product_id	Integer	Unique identifier for the product purchased
category	Categorical	Product category (Electronics, Clothing, Beauty, Home, Books, Toys)
price	Numeric (Float)	Unit price of the product (USD)
quantity	Numeric (Integer)	Number of units ordered
order_date	Date/Time	Timestamp when the order was placed
shipping_date	Date/Time	Timestamp when the order was shipped
delivery_status	Categorical	Current status of the order (Delivered, Shipped, Returned, Pending)

payment_method	Categorical	Payment method used (Credit Card, Debit Card, PayPal, Apple Pay, Google Pay)
device_type	Categorical	Device used to place the order (Desktop, Mobile, Tablet)
channel	Categorical	Marketing channel that drove the order (Organic, Paid Search, Email, Social)
shipping_address	String (Text)	Full shipping address of the customer
billing_address	String (Text)	Full billing address (dropped during cleaning - duplicate info)
customer_segment	Categorical	Customer type (New, Returning, VIP)

During data pre-processing, the following columns were engineered to support deeper analysis:

Engineered Column	Description
total_amount	price x quantity - revenue generated per order
order_month / month_name / month_sort	Month-level breakdown of order_date, with a dedicated sort key so charts display months in correct chronological order
order_year	Year extracted from order_date
order_weekday	Day name (Monday, Tuesday, ...) extracted from order_date
shipping_delay_days	Number of days between order_date and shipping_date

5. Tools & Technologies

- Python: Data cleaning, transformation, feature engineering, and exploratory data analysis, using Pandas, NumPy, Matplotlib, and Seaborn (in place of Excel).
- Power BI: Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Python / Pandas)

Tasks Performed:

- Duplicate Removal: Removed 80 duplicate order rows from the raw 1000-row extract (1000 - > 920 rows).
- Datetime Conversion: Converted order_date and shipping_date from text to proper datetime format.
- Missing Value Treatment: Filled missing categorical values (category, delivery_status, payment_method) with the mode; filled missing numeric values (price, quantity) with the median.
- Duplicate/Unwanted Columns: Dropped the billing_address column since it duplicated information already present in shipping_address.
- Outlier Treatment: Capped extreme outliers in price and quantity using the IQR (Interquartile Range) method instead of dropping rows, to preserve sample size.
- Skewness Check: Verified price and quantity were not heavily skewed after outlier capping, so no log transformation was required.

- Feature Engineering: Created total_amount (revenue), order_month, order_year, order_weekday, shipping_delay_days, and dedicated sort-key columns (month_sort, weekday number) for correct chronological ordering in charts.
- Exploratory Data Analysis: Performed univariate, bivariate, and multivariate analysis using 10 visualizations in Python, each interpreted for business meaning.

Screenshots of Python (Colab) Data Cleaning & EDA Charts:

Order Count by Category, split by delivery status

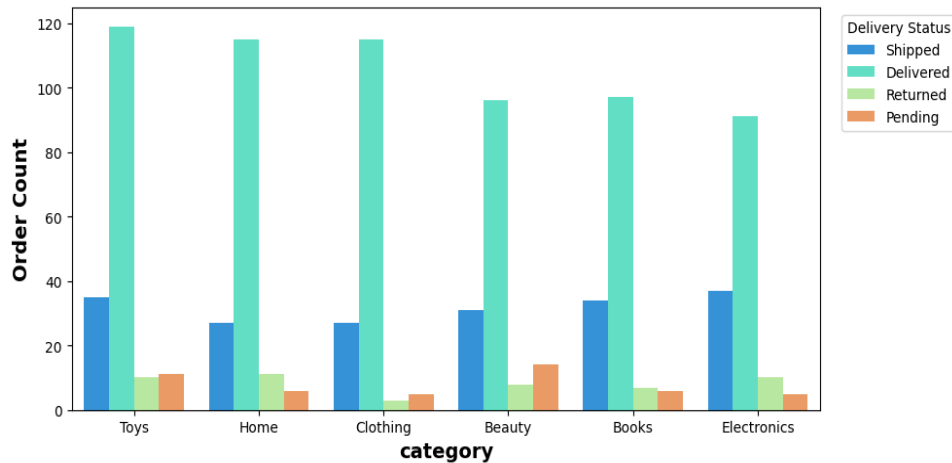


Chart 1: Order Count by Category, split by Delivery Status (Univariate/Bivariate - clustered bar with hue)

Delivery Status Distribution

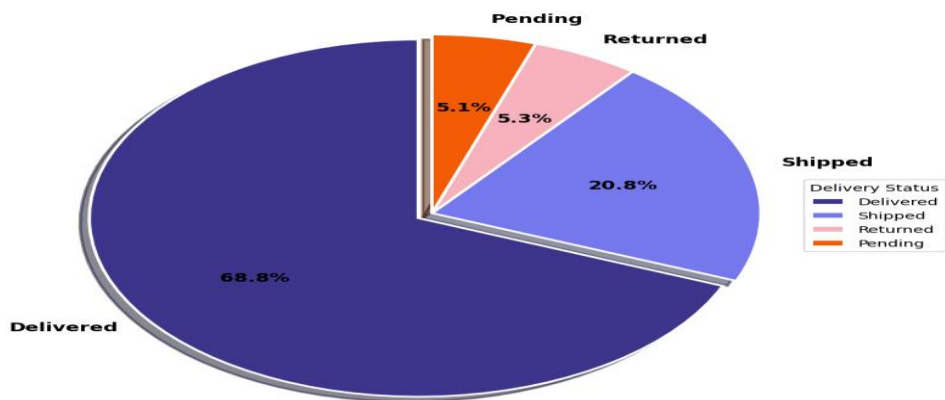


Chart 2: Delivery Status Distribution (Univariate - Pie Chart)

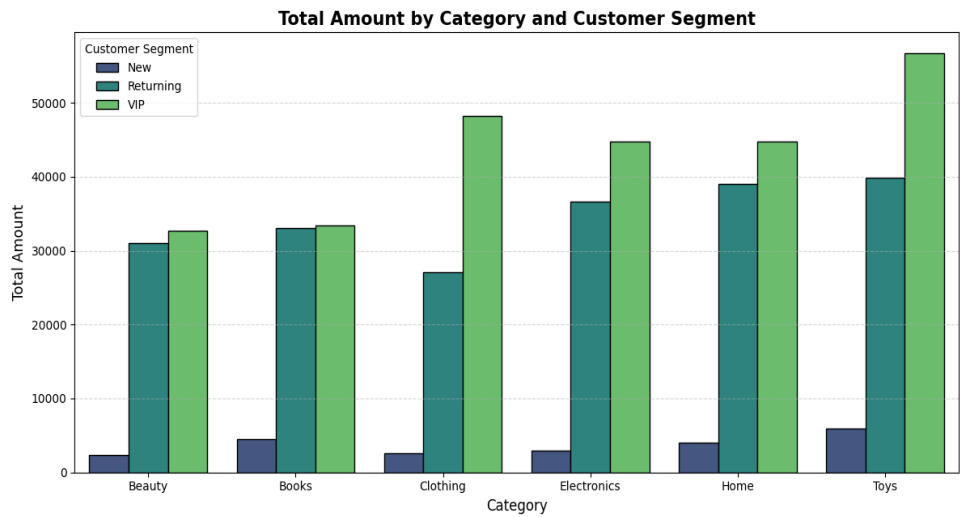


Chart 3: Total Amount by Category and Customer Segment (Multivariate - Clustered Bar via groupby)

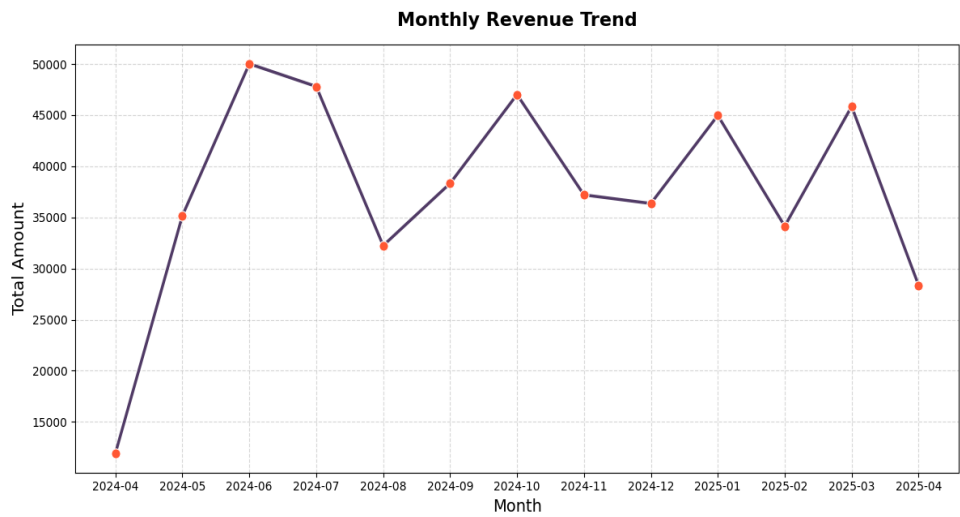


Chart 4: Monthly Revenue Trend (Bivariate - Line Chart, correctly sorted chronologically)

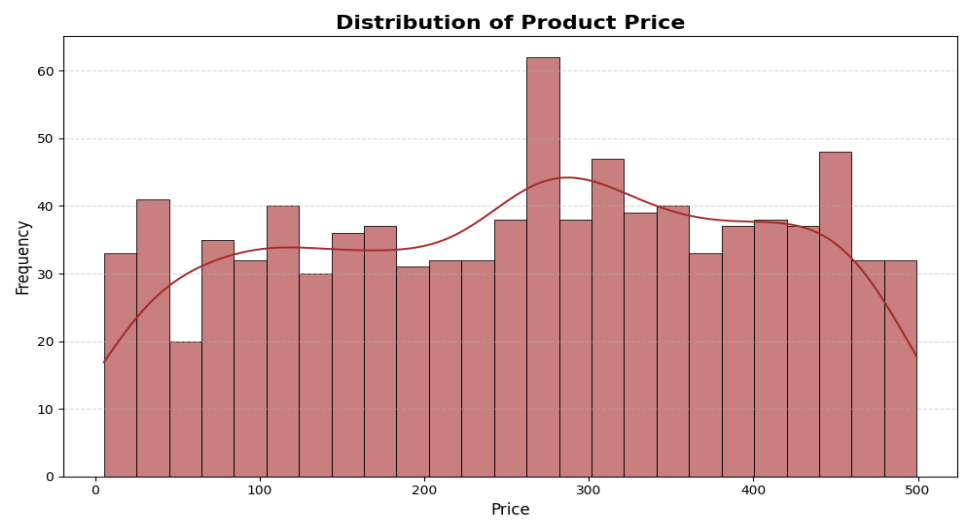


Chart 5: Distribution of Product Price (Univariate - Histogram with KDE)

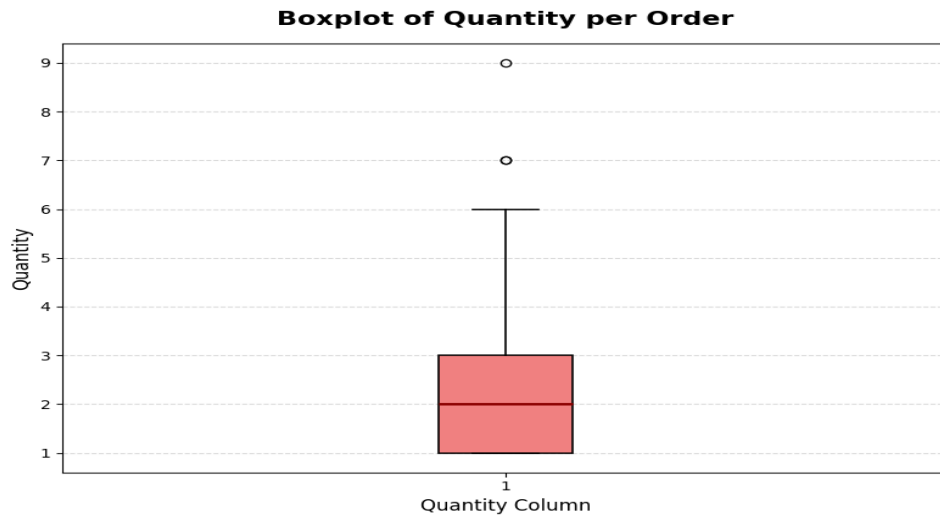


Chart 6: Boxplot of Quantity per Order - Before Outlier Treatment (Univariate)

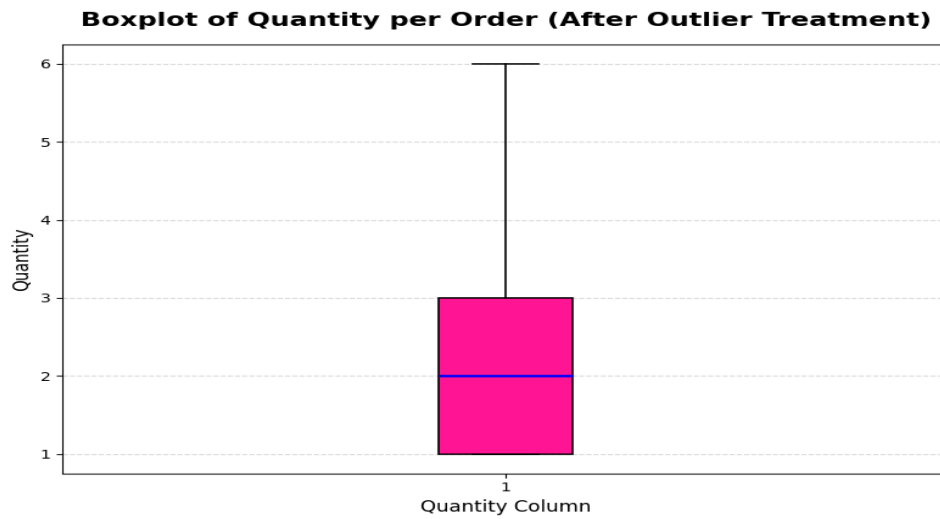


Chart 7: Boxplot of Quantity per Order - After Outlier Treatment (Univariate)

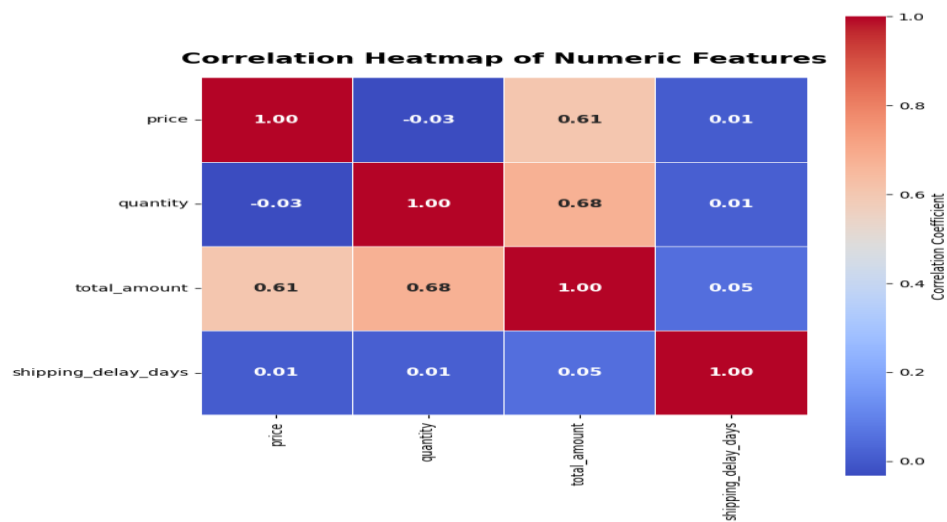


Chart 8: Correlation Heatmap of Numeric Features (Multivariate)



Chart 9: Price vs Quantity, Colored by Category (Multivariate - Scatter Plot)

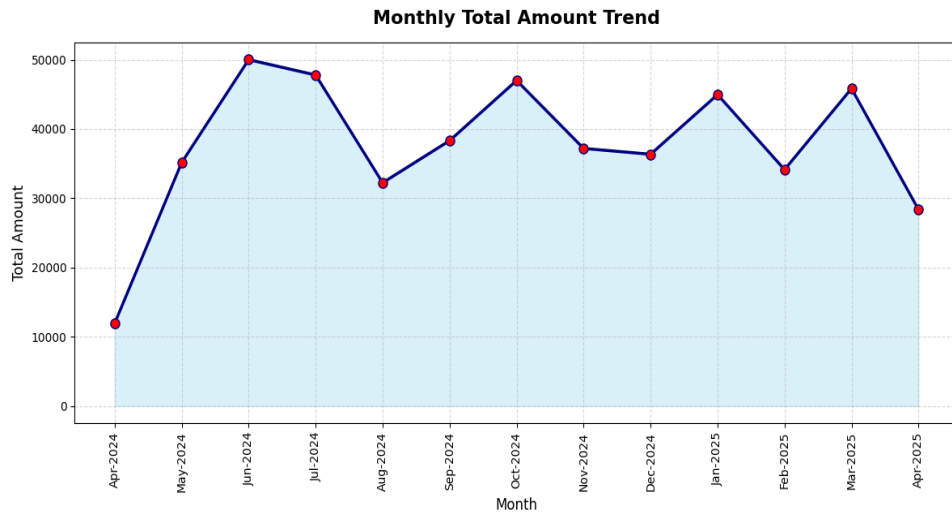


Chart 10: Monthly Total Amount Trend (Bivariate - Area Chart)

Key EDA Observations: Order volume is fairly balanced across all 6 categories. Roughly 68.8% of orders are Delivered, ~20.8% Shipped, ~5.3% Returned, and ~5.1% Pending. Prices are spread fairly evenly between \$5 and \$500. Most orders contain 1-3 units. Total amount correlates strongly with quantity (0.68) and price (0.61), while price and quantity are almost uncorrelated with each other (-0.03), and shipping delay is independent of order size (correlation close to 0).

7. Data Modelling and DAX (Power BI)

Data Model:

- The cleaned dataset (ecommerce_orders_cleaned.csv, exported from the Python notebook using df_clean.to_csv()) was loaded into Power BI via Get Data -> Text/CSV.
- Since the dataset is a single flat transactional table (one row per order), a single-table model was used - no separate fact/dimension tables were required.
- Column data types were corrected in Power Query: order_date and shipping_date set to Date, price/total_amount set to Decimal Number, quantity/order_year/shipping_delay_days set to Whole Number, and remaining columns set to Text.
- A dedicated month_sort column (already engineered in Python) was used with 'Sort by Column' so that month_name displays in correct chronological order (Apr-2024 ... Apr-2025) instead of alphabetical order.
- A custom weekday_num column was added in Power Query (Date.DayOfWeek) and used to sort order_weekday from Monday to Sunday.

Calculated Columns & DAX Measures:

The following DAX measures were created (Home -> New Measure) to power the KPI cards and ratio-based visuals across both dashboards:

Measure Name	DAX Formula
Total Revenue	SUM('ecommerce_orders_cleaned'[total_amount])
Total Orders	DISTINCTCOUNT('ecommerce_orders_cleaned'[order_id])
Average Order Value	DIVIDE([Total Revenue], [Total Orders])
Total Customers	DISTINCTCOUNT('ecommerce_orders_cleaned'[customer_id])

8. Analysis and Visualizations (Power BI)

Two consolidated, interactive dashboard pages were built in Power BI, each with a clear title, labelled axes, legends, and slicers for drill-down filtering.

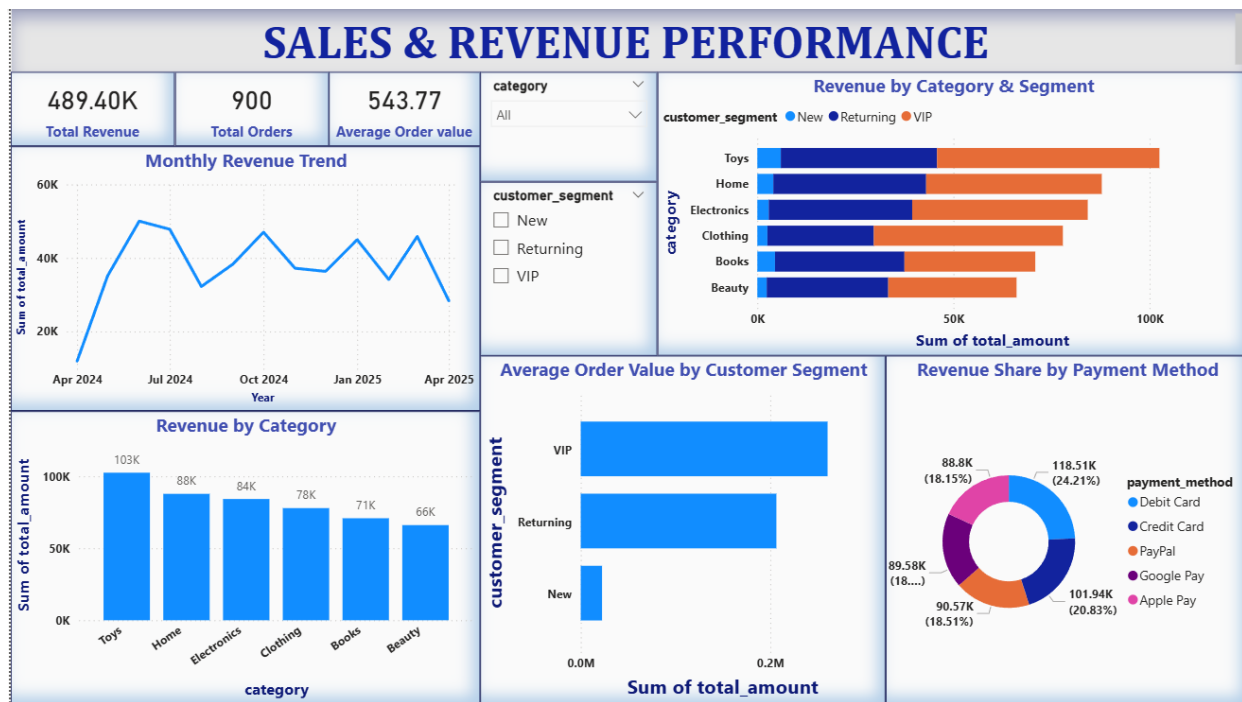
Dashboard 1: Sales & Revenue Performance

This dashboard focuses on overall revenue, category performance, and customer-segment value.

Visual	Chart Type	X-Axis / Category	Y-Axis / Values
Total Revenue	Card	-	[Total Revenue]
Total Orders	Card	-	[Total Orders]
Average Order Value	Card	-	[Average Order Value]
Monthly Revenue Trend	Line Chart	month_name (sorted)	Sum of total_amount
Revenue by Category	Clustered Column	category	Sum of total_amount
Revenue by Category & Segment	Stacked Bar	category (legend: customer_segment)	Sum of total_amount
Average Order Value by Customer Segment	Clustered Bar	customer_segment	Average of total_amount

Revenue Share by Payment Method	Donut Chart	payment_method (legend)	Sum of total_amount
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- Slicers: category, customer_segment (for interactive drill-down/filtering)



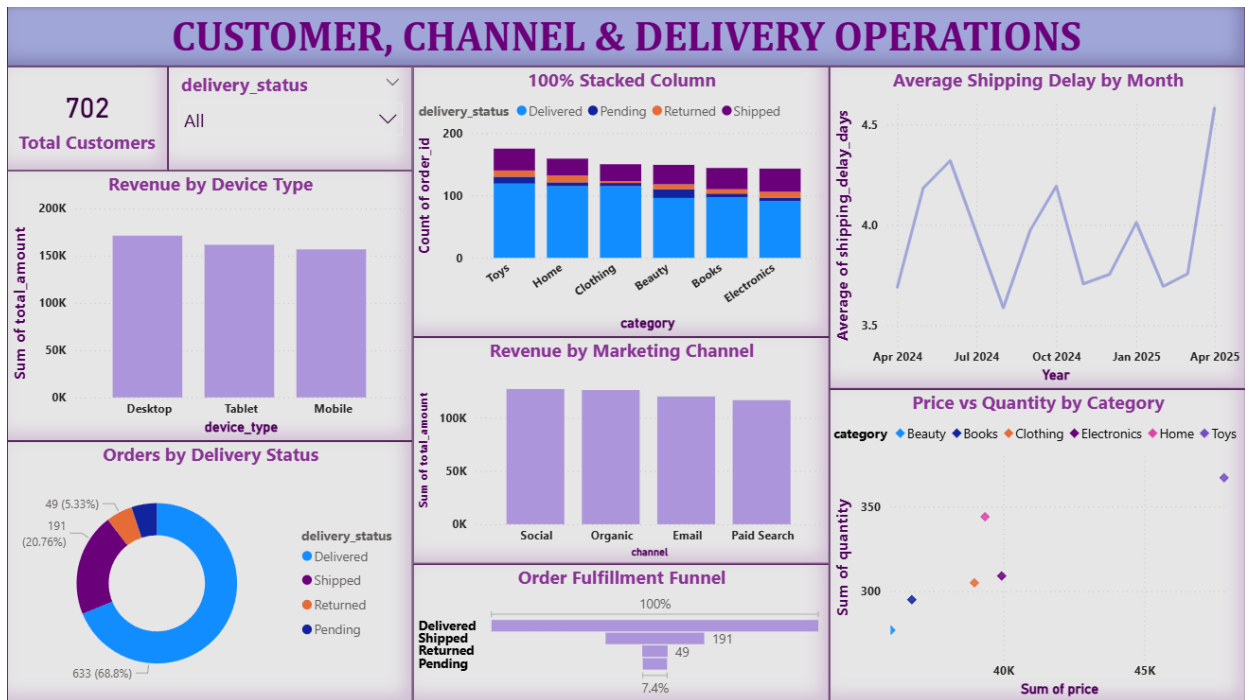
Screenshot: Dashboard 1 - Sales & Revenue Performance

Dashboard 2: Customer, Channel & Delivery Operations

This dashboard focuses on customer reach, marketing channel/device performance, and delivery/fulfillment health.

Visual	Chart Type	X-Axis / Category	Y-Axis / Values
Total Customers	Card	-	[Total Customers]
Orders by Delivery Status	Donut Chart	delivery_status (legend)	Count of order_id
Delivery Status by Category (100%)	100% Stacked Column	category (legend: delivery_status)	Count of order_id
Revenue by Marketing Channel	Clustered Column	channel	Sum of total_amount
Order Fulfillment Funnel	Funnel Chart	delivery_status	Count of order_id
Average Shipping Delay by Month	Line Chart	month_name (sorted)	Average of shipping_delay_days
Revenue by Device Type	Clustered Column	device_type	Sum of total_amount
Price vs Quantity by Category	Scatter Chart	price	quantity (legend: category)

- Slicer: delivery_status (for interactive drill-down/filtering)



Screenshot: Dashboard 2 - Customer, Channel & Delivery Operations

9. Insights & Conclusions

Descriptive Analysis (What is happening):

- Total Revenue across the cleaned dataset is approximately \$489.40K from 900 qualifying orders, at an Average Order Value of \$543.77.
- 702 unique customers placed orders; order volume is fairly balanced across all 6 product categories.
- 68.8% of orders are Delivered, 20.8% Shipped (in transit), 5.3% Returned, and 5.1% Pending.
- Product prices are spread fairly evenly between \$5 and \$500, and most orders contain just 1-3 units.

Diagnostic Analysis (Why it is happening):

- Toys (\$102.5K) and Home (\$87.8K) generate the highest revenue, while Beauty (\$66.1K) trails despite a similar order count - indicating Beauty items have a lower price/quantity per order rather than lower demand.
- VIP and Returning customers have a meaningfully higher Average Order Value than New customers, showing loyal customers are disproportionately valuable.
- Total amount correlates strongly with quantity (0.68) and price (0.61), confirming both genuinely drive revenue, while price and quantity are nearly independent of each other (-0.03) - customers don't necessarily buy more of cheaper items.
- Shipping delay shows almost no correlation with order size, meaning larger orders are not causing logistics slowdowns.

Predictive Analysis (What could happen next):

- Monthly revenue is seasonal rather than steadily trending - it peaked around June 2024 and dipped in April/August, suggesting a recurring demand cycle. While no formal forecasting model was built in this project, this pattern indicates that a time-series forecast (e.g., ARIMA or Prophet) could reasonably predict similar peaks/dips in the next cycle.
- The consistent ~5% Returned rate across categories suggests future return volumes can be estimated proportionally to sales volume for planning reverse-logistics capacity.

Prescriptive Analysis (What should be done):

- Invest more marketing spend in Toys and Home, and introduce bundle offers on Beauty products to lift its average order value.
- Launch a first-purchase incentive (discount or free-shipping voucher) targeted at New customers to raise their AOV closer to Returning/VIP levels.
- Plan inventory and staffing ahead of the June-type seasonal peak identified in the monthly revenue trend.
- Continue investing in the best-performing marketing channels while monitoring Paid Search ROI against Organic/Social.
- Investigate categories or payment methods with a marginally higher Returned share to identify and resolve quality or fit issues.

10. Conclusion

The integration of Python and Power BI proved effective for end-to-end data analysis, from raw, messy order-level data to a fully interactive, decision-ready dashboard report. Python was used to clean, validate, and enrich the raw data and to perform detailed exploratory analysis, while Power BI was used

to build two consolidated, interactive dashboards with DAX-driven KPIs, slicers, and clearly labelled visuals.

The analysis identified which categories and customer segments drive the most revenue, how sales trend seasonally over time, and how delivery performance varies across categories and channels. Together, these findings provide a strong, data-backed foundation for business decisions around marketing spend, customer retention, and inventory planning.